

Applying feminist HCI to ultra-personalization: the case of electric breast pumps

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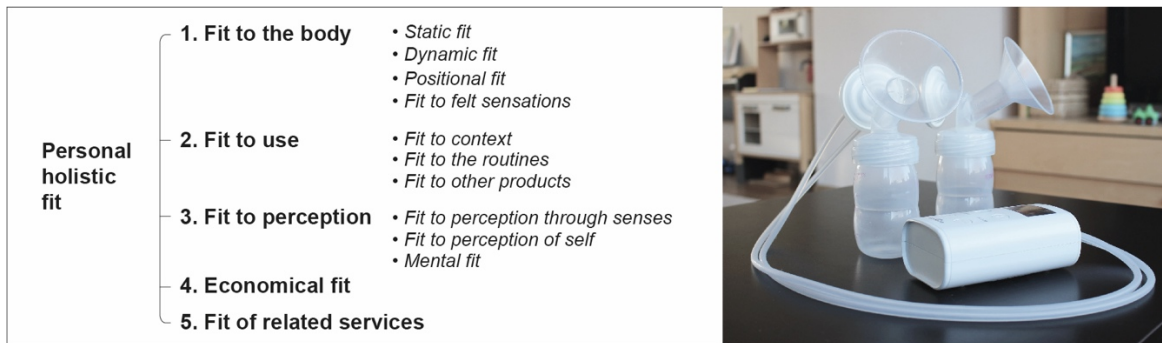


Figure 1: Dimensions and subdimensions of personal holistic fit of electric breast pumps

ABSTRACT

Next to digital products, also physical products can be personalized based on user data collected through digital services and produced by digital manufacturing. This approach is known as ultra-personalization. The ambition of ultra-personalization is to offer products that fit the individual needs of people who will use/wear them. While most of the examples of ultra-personalized products focus on anthropometric fit, in the case of breast pumps the unmet needs go beyond such fit. Being informed by feminist HCI we have conducted interviews with eleven lactating persons and two lactation specialists to gather accounts of embodied situated experiences with currently available electric breast pumps to understand how they fit and misfit users. We have identified five dimensions and ten subdimensions of personal holistic fit. We discuss that designing ultra-personalized products for these dimensions corresponds to feminist HCI values and qualities of pluralism, diversity, embodiment, and self-disclosure.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Ultra-personalization, Feminist HCI qualities, Fit dimensions, Personal holistic fit, Electric breast pumps

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1 INTRODUCTION

We see a growing number of physical and wearable products that are made based on the user's personal data by means of digital fabrication [e.g., 23, 55, 29, 32]. This approach is called ultra-personalization [34] and its ambition is to offer mass-produced products that meet individual user needs and wishes. Most of ultra-personalized product design, is focusing on anthropometric fit. This is much needed as the book 'Invisible Women' by Caroline Perez [38] shows many examples in which the absence of data on women affects the design of apparently gender-neutral products. We were curious how well a gendered product is designed with the aim to find insights for ultra-personalization of product fit.

We took the case of breast pumps¹ to explore how currently available electric breast pumps fit or misfit their users. Literature related to electric breast pumps has brought forward many issues with breast pumps, such as discomfort and pain [10, 14, 26], embarrassment with the sound [26], and perceived effort of routines around breast pump maintenance [26]. These issues suggest that the anthropometric fit of electric breast pumps is not enough to meet

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¹In this case study we use the words "breast milk", "breast pumps", "breastfeeding" and "breast pumping" which is a language that our participants used. People can also chest feed, produce chest milk, and chest pump.

current needs and a more nuanced understanding of personalized product fit is needed.

Breast pumps are considered feminist technology [4], as they expand a lactating person's mobility and choices in infant nutrition and empower women when breastfeeding is challenging (e.g., it is painful, or cannot be done directly in the case of premature babies). At the same time the known issues with breast pumps identify that feminist HCI qualities [1] are not well addressed with currently available electric breast pumps.

To understand the embodied situated experiences of lactating people we have interviewed eleven lactating persons and two lactation specialists. We found that the fit of electric breast pumps has five dimensions, including anthropometric fit, and ten subdimensions that are interrelated. The design of personal holistic fit relies on finding the right combination within these dimensions and subdimensions.

The contribution of this work is threefold. First, we found that experiences with currently available electric breast pumps often do not align with feminist values while they have potential for that. Second, we identified a series of dimensions for a personal holistic fit that is beyond anthropometric fit (Figure 1). Third, we propose that designing for a personal holistic fit can align ultra-personalization with feminist HCI values and qualities.

2 BACKGROUND AND RELATED WORK

2.1 Ultra-personalization

The development of digital technologies, such as 3D scanning and cloud services, allows gathering extensive sets of diverse personal data. Personal data together with digital manufacturing allow multi-stakeholder interaction and product personalization to the level of an individual user. This mass personalization of products is known as "ultra-personalization" and is defined as "*a multi-stakeholder Product Service System that employs capabilities of digital fabrication and users' personal data to produce tangible products and intangible services that together can satisfy individual users' needs*" [34]. We use this term due to the lack of clear definitions for personalization and customization [37] and its focus on user data, digital production, product and service systems, and stakeholder collaboration.

Most of the related work on products that we consider as (potentially) ultra-personalized focuses on creating products based on static and dynamic anthropometric user data or, in other words, based on dimensional measurements of the human body at rest and when moving [39]. The intention of anthropometric fit is to offer future product users and wearers improved comfort, for example, of individualized prosthetics [23], orthopedic casts [55], and 3D printed shoes [35]. Another important offer of the anthropometric fit is improved safety for instance of safety of helmets [24] as well as health solutions such as ultra-personalized tooth implants [31]. Moreover, anthropometric fit allows for improved aesthetics, for example, sunglasses that are designed to fit the face dimensions of a wearer [56]. For garments with integrated electrical components (e.g., vibration motors, stretch sensors) anthropometric fit allows a correct placement of sensor and actuator elements to allow personalized therapy treatments [46].

Apart from the anthropometric fit [46] discusses the fit of the material properties to allow different interactivity possibilities and fit of the programming interactions suggesting a diversity of ways in which ultra-personalized products can fit wearers. While works in ultra-personalization talk about the importance of involving end users in the creation process of ultra-personalized products "to make sure the products fit their individual preferences" [e.g., 45, 46] there are few examples of how users can be involved in the design process.

2.2 Electric breast pumps

An electric-powered (battery-powered or via an electrical outlet, we refer to them as electric breast pumps) breast pump is a promising technology for ultra-personalization. This technology allows lactating people to express breast milk (that is the preferred way to feed infants [53]) and occasionally or periodically be away from their children while continuing to breastfeed. Other reasons for milk expression are difficulties in establishing breastfeeding, premature or low birth weight of infants, feeling unwell, or pain and difficulty with attachment to the breast [19]. A longitudinal survey in the United States revealed that 92% of breastfeeding participants have expressed milk at some point mostly by using a breast pump [25], and according to several works electric breast pumps are the most popular choice compared to manual breast pumps or hand expression [10, 25, 27]. Breast pumps make it easier for the lactating person to express milk and it increases the duration of the breastfeeding of a baby meaning that children receive more human milk than they would if their parents did not use a pump [52].

Along with the advantages research findings indicated significant problems with breast pumps. Comfort, experience, and usability issues may cause people to end breastfeeding earlier than they planned [3]. Participants report discomfort or pain from using pumps [10, 14, 26]. Pumping milk is perceived as a time-consuming and labor-intensive process [12, 26] and usability issues such as setting up and cleaning all the parts [26]. Another concern is the embarrassment of the sound which affects the desire for a discreet and private breast pump experience [26].

It is important to mention that the use of breast pumps is situated in often very physically and emotionally challenging context of breastfeeding [20]. People indicate such experiences associated with breastfeeding as being "*overwhelmed by the enormous responsibility associated with breastfeeding, concerned about failing, and even guilty when they acknowledged their own physical limitations or personal needs*" [20]. Moreover, a negative breastfeeding experience may lead to depressive symptoms in the postpartum phase [7].

The earlier reported issues with pumps show that various needs and wishes of lactating people remain unmet. While more work is needed to explore to what extent the anthropometric fit of the pumps solves the mentioned issues, such as discomfort and pain, there are other needs that cannot be addressed with the anthropometric fit, for example, discreet and private pumping experience and usability issues. Given the importance of breastfeeding for infants and parents, the amount of emotional and physical effort that it requires, and various issues with currently available breast

pumps, this case makes it relevant and necessary to study within the ultra-personalization.

2.3 Notion of product “fit”

Product fit describes how well a product suits the needs and wishes of a (future) user [51]. Mass-produced products are developed in response to the average needs of the largest segment of users from a specific target market and more unique needs of other users remain unanswered [42]. On the other hand, once the future user takes part in the creation of a product, by using digital tools [49], such products offer a much closer fit between the individual needs and product characteristics [42]. In work related to product mass-customization such needs are described as functional needs (a need to achieve an intended outcome [42]), need for uniqueness (“*an individual’s pursuit of differentness relative to others*” [47]), hedonic needs (joy and entertainment from the experience of designing the product [13]), need for self-expression (“*willingness to own a product that fits self-image*” [18]).

For products worn on the body, for example, clothing and wearable technologies, fit is often associated with the way the product conforms to the body of a wearer. Wearers’ needs and wishes around fit are diverse and relate to, for example, personal comfort preferences [48], aesthetics [40], fashion trends and lifestyle [6], and own body image [40]. There is no one definition of “fit” [11, 43, 44], but in essence fit is “*an optimized relationship between human and the environment*” [9]. In clothing, fit is the relationship between the human body and the clothing [43] and the relation and the decision of what constitutes a good fit (in terms of volume, proportion, and comfort) is made by each individual [11]. Shin and colleagues [43] found that the understanding of garment fit differs between professionals and non-professionals. At the same time understanding fit perception from the perspective of non-professionals is necessary to better meet fit needs and desires of future wearers. Several works proposed to develop dimensions of fit to build a deeper understanding of how products worn on bodies fit an individual wearer to better respond to individual needs and wishes. In work related to exosystems (exoskeletons and exosuits) Stirling and colleagues [44] differentiate between static fit, dynamic fit, and cognitive fit that complement each other. Shin and colleagues [43] proposed three dimensions of garment fit from the perspective of non-professionals – physical, functional, and aesthetic fit that are interrelated and grounded in the social context.

2.4 Feminist HCI qualities

We strongly believe in values central to commitments of feminism articulated by Shaowen Bardzell - *agency, fulfillment, identity and the self, equity, empowerment, diversity, and social justice* [1]. She introduced a set of qualities that characterize feminist interaction - *pluralism, participation, advocacy, ecology, embodiment, and self-disclosure*. A combination of these qualities aims to integrate feminism into interaction design research and practice.

Our work is particularly informed by four qualities. The first quality, *pluralism* refers to “designing artifacts that resist a single, universal point of view as [1]. There are “many users, many needs, many voices” and we need to understand a diversity of users

and their needs and explore ways of combining their voices and needs into a solution framework [33]. Moreover, it is important to be sensitive to marginal and marginalized users, to produce more inclusive designs [8]. The second quality, *participation*, refers to engaging and valuing in participatory processes that lead to the creation and evaluation of the designs and promote respect for expertise and different perspectives regardless of status and background [1]. The third, *embodiment*, states that all knowledge is situated in a particular social-cultural and material context [16]. Following the third HCI paradigm interactions with the products are phenomenologically situated, and the goal of the integrations is to support situated action in the world [17]. Feminism puts emphasis on the importance of every type of human experience and subjectively felt life [2]. The fourth quality is *self-disclosure*, which calls for designs to make visible the ways that it affects users and raise awareness of what design is trying to make of them. This creates opportunities for users to define themselves for design.

Building on these four qualities we intended to understand diverse experiences of how currently available electric breast pumps fit or misfit users by directly asking lactating people who use and who do not use this technology. We paid attention to the diverse situated, embodied experiences, and stories of how lactating people perceived pumps and were affected by them.

Breast pumps can be considered feminist technology as they expand a lactating person’s mobility and choices in infant nutrition [4] and empower people when breastfeeding is challenging (e.g., when it is painful, or when premature babies cannot breastfeed directly on the breast). At the same time issues with breast pumps identify that feminist HCI qualities are not well addressed with currently available electric breast pumps.

It is important to mention constraints to the liberatory potential of pumps such as often lack of appropriate spaces at work and time for milk expression [4]. Although there is a desire for smaller, portable, quieter breast pumps [26], [54] argues that making breast pumps smaller, quieter and less visible can reinforce societal norms that make intimate care at work difficult. Our goal in designing better fitting (and hopefully more effective) breast pumps is not a reason to use them as a workaround to challenges around intimate care at work or against “*efforts to fight for longer maternity leaves (and paternity leaves) or the provision of on-site childcare*” [4].

3 USER STUDY

To capture diverse situated, embodied experiences related to the use of currently available electric breast pumps we conducted a series of individual semi-structured interviews with participants with different experiences in using electric breast pumps. The participants identified themselves as mothers and women. We also interviewed two lactation specialists about their extensive observations of how electric breast pumps are being used. Brief position statement: all interviews were carried out by the first author who has experience in using two electric breast pumps at home, work, and other contexts and who faced various frustrating experiences related to the use of electric breast pumps.

Table 1: Overview of participants and pumps they use

Users		Pseudonym	Types of pumps	Number of pumps using/used	Duration of pump use (estimated)
Regular users	1	Laura	Battery-powered	2	12 months
	2	Maya	Double*, Battery-powered	1	4,5 months
	3	Emma	Single*, Double, Battery-powered, Electric, Manual	7	36 months
	4	Zoe	Double, Electric	1	3 months
	5	Min	Double, Battery-powered	1	14 months
	6	Liliana	Double, Battery-powered	1	10 months
	7	Natalie	Double, Electric, Battery-powered	2	24 months
Users for who pump did not work	8	Julia	Single, double, both battery-powered	2 (one did not work)	10 months
	9	Lucia	Double, Battery-powered	1	3 months
Non-users (brief experience with pumps, mostly breastfed)	10	Caroline	Double, Battery-powered	1	1 week
	11	Iris	Manual	1	2 times

*A single pump extracts milk from one breast at a time and double from both breasts at the same time.

3.1 Participants

The study included thirteen participants, two lactation specialists with 13 (pseudonym Anna-LS) and 15 (pseudonym Kim-LS) years of experience, and eleven lactating persons (Table 1). We organized two recruitment rounds to find people willing to participate. We first placed advertisements on the social media channels of the local expat community. In the invitation, we asked participants to share their experiences with the electric breast pumps they used. We found that for seven of them who initially joined electric breast pumps generally worked. To explore more diverse perspectives on electric breast pumps we used the same social media groups and invited participants for whom electric breast pumps did not work or who never used electric pumps. Four lactating persons joined from whom Caroline and Iris had only a brief experience with electric breast pumps and mostly fed at the breast and Julia with Lucia had a very frustrating experience that caused them to either nearly (Julia) or completely (Lucia) stop breastfeeding. Julia had a good experience with her second pump and wanted to share her experience with the first pump that caused her to nearly terminate breastfeeding.

Only English-speaking lactating people, with (self-perceived by participants as) uncomplicated pregnancies were invited. The study was reviewed and approved by the University ethics committee and all participants signed informed consent forms.

3.2 Set-up

During semi-structured individual interviews, lasting 35 min to 2,5 hours, participants were asked about their experiences with the electric pumps they use, and lactation specialists were asked about lactating persons' experiences they had observed during their practice as lactation consultants. We first asked participants general questions about the pumps they use (type, history of use, acquisition method – buying/getting second hand/renting). Then

we asked about their perception of the electric pumps they used and how they think their pumps fit or did not fit them in their daily lives. After that we asked more focused questions about how electric breast pumps fit or misfit the body, initial expectations from the pump, routines related to the use of the pump, what participants liked and disliked about their electric pumps, whether there were any complaints, whether participants adopted any workarounds when using pumps, whether perceptions of other people influenced the experiences related to the pumps.

3.3 Data analysis

All interviews resulted in rich data consisting of nearly 13 hours of audio recordings. Audio recordings were transcribed and analyzed by the first author using reflective thematic analysis [5]. The first author read and reread the transcripts to familiarize with the data and made notes. The first author then explored and developed patterns across the dataset and iteratively developed codes that evolved through the coding process. After that, the first author constructed the initial themes that were discussed over multiple meetings with the second and the third authors and we collaboratively developed higher-level themes and gave them clearer names.

4 FINDINGS

Our analysis resulted in five fit dimensions and ten subdimensions (Table 2) describing how the electric breast pumps can fit an individual user.

4.1 Fit to the body

Part of the pump – breast shields – are placed directly over the breasts and a good fit between the breast/nipple and shields is necessary for the comfort and the efficiency of the breast pumping. Shields are selected based on the diameter of the nipple and once the vacuum is applied the nipple should be centered in the shield tunnel

Table 2: Dimensions and subdimensions of personal holistic fit

	Fit dimension	Fit subdimension	Description
Personal holistic fit	1 Fit to the body	1.1 Static fit	Fit between the user's body shape and the product
		1.2 Dynamic fit	Dynamic relation between the user's body and the product
		1.3 Positional fit	Desired position on the body for the effectiveness of the use
		1.4 Fit to felt sensations	Fit to felt sensations from the product use
	2 Fit to use	2.1 Fit to context	Fit to physical and social contexts
		2.2 Fit to the routines	Fit to routines associated with the use
		2.3 Fit to other products	Fit between the product and other products that a user wears/uses
	3 Fit to perception	3.1 Fit to perception through senses	Fit to visual, auditory, olfactory, tactile, and gustatory aspects of the product
		3.2 Fit to perception of self	Fit between the product and the self-image of a user
		3.3 Mental fit	Fit between the product use and intentions of a user
	4 Economical fit		The fit between product price and the user's willingness to pay for it
	5 Fit of related services		Fit of services related to designing/buying/using the product

and move freely during the pumping without causing discomfort. Our findings suggest that lactating people use shields of the wrong sizes, have issues with the vacuum, need to learn how to position the shields properly, and often experience pain while using pumps.

4.1.1 Static fit. Both lactation specialists observed that people they consult often use the wrong size shields. One of them said: *"I am not a number person but two out of five [women use wrong size shields]"*. During interviews, participants talked about three main reasons for the misfit of the shields – lack of awareness about different sizes, difficulties with fit assessment, and fit change over time.

Seven participants said that they were not aware of the existence of different shield sizes. For example, Laura, mother of three children, who have experience with two pumps, was surprised to hear that shield have different sizes: *"I had no clue about sizes, I am pretty much M size, so I guessed they [shields] should be M. And they [shields] were OK. And I never really measured myself, I did not even know you could measure. I would not even know what you need to measure. I never really took the time to investigate which shields I had. Well, they do not fit so well, maybe I should look into that"*. While Laura achieved her breastfeeding goals, Lucia, a first-time mother, who also was not aware of different shield sizes, told me that her overall breast pumping experience was very negative and painful possibly because of the wrong fit, and that caused her to stop breastfeeding: *"It was a very painful and negative experience for about 3 months in total [...] Maybe I did not have the right size because I don't have big boobs in general, so I'm not sure if I had the wrong size [...]. I thought they just fit everyone, I did not measure them. No, I didn't even know"*.

Assessing fit can be challenging for a lactating person as well as for healthcare workers. Kim-LS encounters wrong recommendations by nurses who attend to parents after childbirth: *"Yesterday there was a woman with small nipples and 27 mm breast shield. The nurse recommended it"*. For Liliana, a mother of two, with ten months of experience in using the pump and who initially started to use the breast pump to increase lactation after childbirth, assessing

the fit of the pump is still challenging: *"It was not easy [to assess the fit of the shield]. I think it took me quite some time to research on the Internet trying to find videos or some pictures on what was the right size of the flange compared to my nipple size. And still not clear"*.

The size of that flange can no longer fit over time. Kim-LS mentioned: *"It is also possible that on the first day [size] 21 fits well and then [size] 24 fits better"*. Liliana observed that her breasts change during the breast pumping session: *"The breast is changing a lot during breastfeeding session not only the nipple is changing in size, but also the breast itself"*.

4.1.2 Dynamic fit. Seven participants talked about the fit between shields and breasts when the vacuum of a pump is applied, in particular, they talked about the fit and misfit of vacuum power, misfit of the shield materials, and changes in fit over time.

One of the reasons participants were not satisfied with the pumps they used was the power of the vacuum. The vacuum power of Min's pump is too low for her and as a workaround she uses only one shield at a time, doubling her pump session time: *"I have a double one. This is very weak, so I like [to use only one side at a time]. It will take one hour [to pump the milk]"*. For Emma, the first pump she ever used was too powerful and she had to find a replacement: *"It was horrible because it had only one speed and it was very powerful"*.

Lactation specialists and participants find it important that the material of the shield fits the person and preferences for material differ. Anna-LS said: *"Some women like the pillow [cushion insert in the shield] in it and others without the pillow [cushion]"*. The material that does not fit can cause damage according to Kim-LS: *"Some people can even get blisters, burns [from soft cushions]"*. For Julia, the unfitting shield caused her a lot of pain during the pumping, and she was close to terminating the breastfeeding: *"Because the cups [shields] were [from hard] plastic they were not working [well] with breasts, yeah, it wasn't a nice feeling in the end, and I was very close to giving up on breastfeeding"*.

The fit between the shields, breasts, and vacuum is applied changes over time. Liliana found that the shields fit her differently at the beginning and at the end of the pumping session: *“At the beginning of the pumping session, it doesn’t fit [the shield is too small], and then by the end of the pumping session [...] I have softer breasts then you get the feeling like this [shield] may be too big for me”*. Emma, who used a breast pump until her son was 3 years old, commented on the overall fit of the vacuum over time: *“During the first week electric breast pump, if it gets a bit of ejection reflex, it will do the thing. When pumping for a child of 3-4 years of age it is a vastly different pumping experience because the breast doesn’t let itself get fooled anymore [the ejection reflex is harder to achieve]”*.

4.1.3 Positional fit. The way breast shields are positioned on the breast influences the effectiveness of the breast pumping [15]. The shields must be positioned over the breasts so that the nipple is centered in the tunnel of the shield and the shields do not put much pressure on the breasts not to obstruct milk flow. Positioning and maintaining the position can be challenging according to two lactation specialists. Anna-LS said *“Holding the shield can be problematic for some women, they don’t hold them in the proper way and the field is not fitting well”*. Kim-SA added that wrong position of shields can cause problems: *“If you are holding in a way that you block your milk ducts this effects milk expression. So, you need to hold it but not push. It [pushing] can also result in problems – infections”* (Kim-LS). Laura admitted that it took her time to learn how to hold the shield properly: *“You really try to find yourself a comfortable position that does not hurt you and that the milk comes in here [points at the funnel of the shield]”*.

4.1.4 Fit to felt sensations. Discomfort and pain reduce the oxytocin reflex (also known as the let-down reflex) and this inhibits the flow of the milk [36]. It is known that lactating people experience and report feeling discomfort or pain while breast pumping [10, 26]. In our study, six people experienced pain from using electric breast pumps. For two persons the pain was such that Julia nearly gave up breastfeeding *“I would have stopped breastfeeding around week six really [if my friend would not have given me another pump]”*, and Lucia stopped breastfeeding: *“It was very painful and negative experience for about 3 months and after suffering my milk supply kind of stopped. [...] If I have another baby, I will definitely want to use formula straight away [...] for me this was a very traumatic experience because of the pain it caused me.”*

Interestingly two persons who experienced pain while breast pumping received encouragement from nurses to endure pain and wait till it passed. Laura said: *“It [breast pumping] was really painful and she [the nurse] said that the child is also painful [referring to breastfeeding] so this is normal that it [breast pumping] is painful”*. Maya shared similar advice from the nurse: *“It was so painful [...] but the nurse told me “That is normal. Keep on going, a few more days”. And now it is fine”*. On the contrary, lactation specialists help people to minimize discomfort by trying out and finding shields that are comfortable: *“I listen to a mum and we try different shields “Is this hurting? Use this one. Is it better? Then we use that one”. So I listen to the feelings of a mum”*.

4.2 Fit to use

Context of use has been seen to have a “profound effect” on the use of the product [28]. During interviews, people mentioned multiple examples of how electric pumps fit or misfit context of use (physical and social), fit other pump-related and unrelated products, and fit to routines related to the use of pumps.

4.2.1 Fit to the context of use. The needs around the electric pumps’ fit to the physical context were clearly mentioned by Emma, who referred her to own three years of experience and the experience of other lactating people whom she met while being a local LaLeche leader (a volunteer who supports breastfeeding and provides peer-to-peer support). Emma said: *“I think that a lot of these [pumps] things are made for one certain idea in mind. Being a mom sitting in an office situation with a table. . . Companies have to realize that that’s only one of the many scenarios I know of moms who were taxi drivers that had to pump in their taxi on a taxi stand [...] I know that that in the Netherlands law says you have a right to a pumping room. In practice that is just not the case. A lot of mums pump in like a back room where the archives are, you know, or just the cleaning room or toilets. I’ve pumped in toilets too many times. [...] If I’m in the toilet, or if I’m in a cleaning back closet somewhere and I don’t have a table to put things on, it would be really nice to have the other style [of bag], which I could zip open from my side, and it sort of folds out and there’s a place for the pump and for the parts.”* From the comments of Emma we see an opportunity to design interactions with the pumps that could fit the various physical settings of use contexts.

For three persons the fit to the social context was important. Laura and Natalie, who were using breast pumps to be able to return to work that they liked, admitted that the bags they use to carry their pumps to work look neutral and do not associate with breast pumping, as Natalie mentioned: *“It is a subtle bag, you know, like not like hey “I’m a woman pumping around” you know”*. Maya, who initially started to use the pump to have the freedom to meet her friends and later to be able to go back to work, talked about her embracement of using the pump at work where others [men] could hear her: *“I am working in the army, so I have only men around me [...] am always feeling a bit uncomfortable with the noise”*.

4.2.2 Fit to other products. The decision to purchase certain electric pumps often depends on pump-related products that people already have or plan to get with the pump. Laura and Min have decided to get their pump because of the accessories they had such as bottles and silicone nipples. Moreover, Laura had a very good experience with the pacifier and she expected that the breast pump from the same brand would fit her well too: *“I was really happy with the pacifier [...] And if you have a pacifier then you buy bottles and then you get the breast pump as well”*.

4.2.3 Fit to routines. There is a set of routines that accompany the use of breast pumps. During interviews, lactating people mentioned a number of routines that help them before, during, and after breast pumping. Interesting that people have various routines that fit them and allow them to produce enough/more milk. For Min it helps to *“drink a lot of warm water, keep the feet warm and just relaxed”*, for Liliana it helps to *“drink a lot of water and if I get a*

lot of carbohydrates”, for Emma a sound of her hungry crying baby always helped her when the breast pumping was not going well.

Another set of routines reported by the participants is the time of pumping. For example, Maya and Natalie often do cycles of pumping, for example, 10 minutes pumping, resting, and again 10 min pumping to get enough milk, while Liliana always pumps the same amount of time. Zoe is pumping in the morning after feeding her child to collect more milk for the daycare. The described routines fit our participants and allowed them to achieve their goals. We find that diverse personal strategies around the use of the pumps can inform other people, for example, novice users, and help them in achieving their goals.

When asked how they wished electric pump fit their routines better, six people said they would like to have free hands and move around while pumping. Although wearable pumps are emerging on the market, none of the lactating persons who participated in this study had them.

4.3 Fit to perception

4.3.1 Fit to sensory perception. Sensory impressions acquired through senses (seeing, hearing, touching, tasting, and smelling) contribute to user product experience [42] and people differ extensively in their sensory reactions to products [22]. During interviews, only audio and visual and touch aspects of electric breast pumps were mentioned. We have placed the touch aspect of the pump to the *“fit to the body”* section as it directly relates to the dimensions of the breast shield and vacuum of the breast pump. Five lactating persons said that they were either negatively surprised or annoyed with the sound of their pump. For Min the sound made her feel stressed: *“It is very noisy. When you pump you really have to be relaxed? But you hear that sound [and it is] not relaxing, this is very stressful”* and stress is known to have an adverse effect on the breast milk supply. When it comes to the visual aspect of the pumps two persons found their electric pumps visually appealing and two did not like the look of the shields they use. Emma said regarding the shields: *“If you look at this, it doesn’t elicit any happy feelings [...] it doesn’t look inviting. It’s plastic and clinical”*.

4.3.2 Fit to perception of self. Literature suggests that people have a greater preference for products that are more coinciding with their self-image [29]. Self-image congruence refers to the match between a person’s self-concept and the image (or “personality”) of a given product [21]. In our study, lactation specialists have talked about the pumps and their relation to the person’s self-image. Anna-LS said: *“When I hear something about a pump a mum says: ‘I am like a cow’ and Kim-LS added that such association is not pleasant for people: ‘Some people think ‘Oh, I am a cow’ and a connection with that is not so nice for them’*.

4.3.3 Mental fit. People’s perceptions of breast pumping can influence the overall effectiveness of breast pumping. Anna-LS said: *“I learned one thing when I became the lactation specialist – you are feeding your baby 25% from here [pointing at the right breast], 25% from here [pointing at the left breast], and 50% from here [pointing at the head]”*. Kim-LS shared a similar idea that perceptions of a person influence the process of breastfeeding and breast pumping:

“Breastfeeding you do with your mind, it is a total mind thing. If you think ‘no, not going to do it’ – then you are stuck”.

4.4 Economical fit

The price of the breast pump matters when choosing the type and the model of the breast pump. For Emma the pump was a *“big spending point”* and she bought her first pump *“because it was affordable”*. Iris, who initially got the pump *“just in case”* she needed it in the hospital after the birth, also based her choice of the pump on the price: *“So I didn’t choose for an electric one because it was really expensive”*. Lucia said that it was not clear to her what advantages a more expensive pump would offer her: *“I’ve compromised in terms of price versus quality because we just bought the one that is more affordable than [name of the brand]. Because you’re like ‘OK, why would I buy [name of the brand] which cost like 200 euros if I can buy something for 100 euros?’”*. Moreover, the switching costs when changing pumps and an intended use time can be a barrier to getting a better-fitting breast pump. For example, Laura thought that wearable pumps might fit her better, but she was not prepared to buy them for half year of use: *“I am not going to spend another few hundred euros for a half year of pumping”*.

4.5 Fit of related services

Both lactation specialists said that it is important to first try the pump to understand if it fits. Both lactation specialists used an analogy of shoe fitting, as Kim-LS said: *“Never buy a pump of 300 euros without trying. This is the same about shoes, don’t buy shoes that you want before fitting”*. Min wished to have an opportunity to try different pumps: *“I think if I can wish something that should be a trial of the pump”*.

Most of the participants stressed that they wished to have more information related to the use of breast pumps. Emma, who has a lot of experience with pumps from various brands, said: *“I think that a lot of companies would really help women, feeding people, use their products efficiently by guiding them to help”*. Liliana finds that she learned a lot by using a breast pump and wished she had that knowledge readily available before she started using her breast pump: *“There is so much knowledge with people who are doing it [using breast pumps] but it is [the directions for effective use] actually not written anywhere”*. Lucia, who referred to her breast pumping experience as a big struggle particularly missed necessary information: *“I think the whole information regarding pumps is very scarce. I believe it’s just that most people assume that you would breastfeed [...]. But if that doesn’t work, you always have the pumps of course. But then nobody is giving you extra information that you might need not to hurt yourself with it. Because after all it’s just a tool, but it can be harmful as it was for me.”*

4.6 Personal holistic fit

Both lactation specialists stressed the importance of the holistic fit of the pumps. As Anna-LS said: *“The good fitting helps a lot. And when it feels comfortable it helps a lot. And when it looks good it helps a lot. When it does not make a lot of noise it helps a lot”*. Dimensions of fit are interrelated, for example, there is a relation between static and dynamic fit as Kim-LS put it: *“The size of the shield can be compensated by the amount of the vacuum”*.

5 DISCUSSION

Breast pumps, as a technology have the potential to align with the feminist values of agency, fulfillment, identity of the self and empowerment. Breast pumps support lactating person's agency around decisions towards the breastfeeding goals and this applies to all our participants who are regular users of electric breast pumps. Breast pumps sustain lactating person's feelings of fulfillment, for example, the electric breast pump helped Liliana increase milk production after childbirth and that was something that she desired. Breast pumps empower lactating people to have control over their own lives, for example, Maya initially used her breast pump to be able to have freedom to meet friends without her baby. The increased mobility that breast pumps offer supports person's identity of self as, for example, a "friend" (Maya) and "being a working professional" (Laura and Natalie). However, currently available electric breast pumps still do not meet many needs and wishes of their users. For example, in our study, participants talked about pain when using electric breast pumps (six lactating people), embarrassment (Maya, Emma), and extreme frustration (Julia, Lucia). Moreover, some nurses can give people wrong advice on sizes, and some encourage them to tolerate the pain while pain and stress are well-known factors that inhibit milk supply [36]. The authors question whether pain and frustration are the price lactating people must pay to increase their mobility or feed a premature baby.

From the accounts, we can conclude that electric breast pumps often are not aligned with feminist values. Lactating people lack **agency** when the nurses advise them to endure the pain or when they are unaware of the available sizes of shields. There is an issue with **identity of the self** as according to lactation specialists lactating people negatively perceive themselves as "cows". In the account of Lucia, unsolved issues with her pump caused her to finish lactation much sooner than she wanted which is against the value of **fulfillment** to care for her child. Emma commented that breast pumps are designed with one context in mind "office situation with a table" while the context of possible use is much more diverse, leaving lactating people with less **empowerment**.

The accounts of our participants on the fit of electric breast pumps helped us to identify dimensions and subdimensions of the personal holistic fit. The dimensions and subdimensions of personal holistic fit can be used to analyze the fit of electric breast pumps as well as to inform directions for research and design toward ultra-personalization. A further focus on the feminist qualities within the development of ultra-personalization can help in the following ways.

Our findings highlight a diversity of needs that should inform design of electric breast pumps. We particularly want to capture this diversity to learn about the meaningful differences and relations between fit dimensions to align with the feminist quality of **pluralism**. We foresee that meaningful combinations of various personal needs towards the fit of electric breast pumps will result in pumps that fit individuals differently. In other words, the personal holistic fit of the electric pump will differ per user and there won't be a universal type of product fit. For example, for some persons the design of the pumps should allow them to pump in spaces that are not properly equipped for breast pumping, while for others the noise is the most important factor that affects them (socially -

feeling embarrassed, and physically – sound causes stress that has adverse effect on the breast milk supply).

Accounts, that participants shared, related to their **embodied** situated experiences within their social-cultural and material contexts. Ultra-personalized breast pumps that aim to address the needs and wishes of individual users should therefore support the situated actions of pumps' users.

Capturing individual needs of users is a labor-intensive activity in particular if needs are heterogeneous [49]. One of the approaches for identifying and capturing users' needs is through digital toolkits (also known as toolkits for innovation). Such tools can shift need-related activities to hands of users [50] and empower them to participate in the design of the ultra-personalized product that corresponds to the feminist quality of **participation**. We envision that some of the digital services will inform product design "*before the use*" of the product. For example, 3D scans can inform ultra-personalization of the product shape (*static fit*), and text-to-image AI tools can support future users in the design of the visual appearance of the products (*fit to perception through senses*). Other services can inform product design "*during the use*". For example, for the *fit to felt sensations* users could be provided with specially designed apps where they capture detailed subjective felt experiences of how the product fits or misfits them. Such information then could be used by the company to understand possible issues with the product use and potentially offer a solution in the form of advice or a new ultra-personalized (part of a) product.

Designing with personal data asks users to share their intimate information, for example, 3D body scans. Previous work on body scanning identified various tensions that are associated with capturing, visualizing, and sharing personal body-related data [30]. In line with self-disclosure quality, we see that it is necessary to inform users what personal data is being captured about them and how it will be used for the design of ultra-personalized breast pumps. The same self-disclosure should be done in relation to the translation of personal data to personal fit. This data-personal fit translation should be visible to users so that they can do something about it, for example, to rate it or comment in case of a misfit that will inform the design of a better-fitting electric breast pump.

Several identified dimensions and subdimensions of personal holistic fit relate to earlier definitions of fit. *Static fit* relates to Shin and colleagues' [43] *physical fit* and Stirling and colleagues' *static fit* [44]. *Dynamic fit* relates to Shin and colleagues [43] *functional fit* and Stirling and colleagues' *dynamic fit*. *Fit to perception of self* relates to Shin and colleagues' *aesthetic fit*.

We see two opportunities for further research to expand and/or detail proposed dimensions of personal holistic fit. First, there is a need to gather more situated experiences with breast pumps that occur in different countries since country-specific policies to parental leaves and rights, and public health recommendations may affect the overall breastfeeding and pumping journeys. Lactating people who participated in the study came from different parts of the world (Venezuela, Vietnam, Poland, the US, Bulgaria, and the Netherlands) and had most of their breast-pumping experiences taking place in the Netherlands. Second, in this work we have focused on the users' and lactation specialists' perspectives, who are both non-experts in breast pump design, towards fit and misfit of electric breast pumps. Shin and colleagues [43] argue that it is important to investigate

the perspectives of non-professionals to better meet their needs. Therefore, further work should investigate how experts in breast pump design perceive the proposed personal holistic fit and explore how their professional perspective can develop it further.

6 CONCLUSION

A growing number of works related to product ultra-personalization concentrates mostly on the anthropometric fit. While this focus is much needed, our case of the electric breast pumps show that anthropometric fit is not sufficient to address the range of existing issues. To understand how currently available electric breast pumps fit or misfit users we have gathered accounts of eleven lactating persons and two lactation specialists. The accounts highlight how the values of feminism are underdressed with currently available electric breast pumps. We have identified five dimensions and ten subdimensions of personal holistic fit from which the anthropometric fit is only a part and reflected how designing for personal holistic fit corresponds to feminist HCI qualities. The dimensions of personal holistic fit can be used to analyze the existing electric breast pumps or to inform research and design of ultra-personalized electric breast pumps.

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